

# **Project Proposal 1: LiviSmart – AI-Powered Cost of Living Optimization Platform**

## **1. Introduction**

The escalating cost of living in Ethiopia presents a significant and growing challenge for households, students, and low- to middle-income families. Daily expenses for food, transportation, housing, and utilities are subject to frequent fluctuations and vary considerably across different locations. This financial instability makes it difficult for individuals to budget effectively, make informed purchasing decisions, and maintain a stable standard of living. Many families currently rely on guesswork or informal networks to manage their expenses, often leading to financial stress and inefficient spending.

## **2. Statement of the Problem**

Currently, Ethiopian consumers face a critical lack of price transparency and predictive tools for managing their household finances. Key problems include the absence of a centralized system for comparing real-time prices of essential goods across different neighborhoods and cities, and the reliance on manual, reactive budgeting methods. There is no intelligent mechanism to forecast price trends for items like food and transport, or to provide personalized, data-driven advice on cost-saving alternatives. This information gap forces households into reactive financial decisions, diminishing their purchasing power and contributing to widespread economic anxiety. Policymakers and local vendors also lack aggregated, micro-level data on consumer cost pressures, hindering effective policy and business responses.

## **3. Objectives**

The primary objective of LiviSmart is to develop an AI-powered platform that optimizes household spending. This will be achieved through:

Aggregating and analyzing real-time price data for essential goods and services.

Implementing machine learning models to predict short-term price trends and alert users to potential cost increases.

Creating an intelligent budget planner that provides income-based spending recommendations.

Developing a location-based comparison tool to identify affordable vendors and neighborhoods.

Generating personalized alerts and cost-saving tips based on individual household profiles and spending patterns.

## **4. Significance of the Project**

LiviSmart will empower Ethiopian households to transition from reactive spending to proactive financial management, directly reducing financial stress and improving their quality of life. By promoting price transparency and data-driven decision-making, the platform will foster greater fairness and accountability in local markets. The aggregated cost-of-living data and analytics generated by the system can serve as a valuable resource for economists, researchers, and policymakers to understand inflation dynamics at a granular level. Ultimately, LiviSmart aims to be more than just an expense tracker; it will be a comprehensive, AI-driven financial ally for navigating the complexities of the modern economy.

# **Project Proposal 2: EthioHealth Sentinel – AI-Powered Disease & Mortality Monitoring and Advisory Platform**

## **1. Introduction**

Public health threats such as disease outbreaks often spread rapidly in communities before official responses can be mobilized. In Ethiopia, citizens frequently remain unaware of local health risks until an outbreak has already affected their community. This lack of timely information delays preventive actions, increases the spread of illness, and places unnecessary strain on the healthcare system. A system that monitors health trends and communicates risks directly to the public could significantly improve community-level health outcomes.

## **2. Statement of the Problem**

In Ethiopia, disease outbreak and mortality data are often scattered across health facilities and inaccessible to the general public. Citizens have no reliable way to know whether cases of communicable diseases are increasing in their neighborhood or city. Health authorities may detect trends but lack an efficient mechanism to broadcast timely warnings and preventive guidance to those most at risk. Key challenges include:

- No centralized platform for the public to view local disease trends and mortality data.
- Delayed public awareness of emerging outbreaks, allowing diseases to spread further.
- Absence of automated systems that translate health data into actionable public advisories.
- Limited ability for individuals to receive personalized alerts about health risks in their area.

## **3. Objectives**

The objective of EthioHealth Sentinel is to develop an AI-powered platform that monitors disease and mortality data and provides real-time public health advisories. Specific objectives include:

- Creating a data entry system for health authorities to register disease cases and deaths.
- Developing interactive maps and dashboards for citizens to visualize local health trends.
- Implementing an AI module to detect unusual spikes in cases or deaths automatically.
- Generating automated health advisories with symptoms, prevention tips, and treatment guidance when anomalies are detected.
- Establishing notification systems (SMS, email, or in-app alerts) to inform users of emerging risks in their area.

## **4. Significance of the Project**

EthioHealth Sentinel will bridge the critical information gap between health authorities and the public. By providing citizens with real-time, actionable insights about local health risks, the platform empowers individuals and families to take preventive measures before outbreaks affect them directly. The AI-driven advisory system ensures that warnings are not only timely but also practical, offering clear guidance on protection and response. For health authorities, the platform offers improved visibility into community-level trends and a direct channel for public communication. Ultimately, this project contributes to building a more informed, prepared, and resilient society capable of responding proactively to public health challenges.

# **Project Proposal 3:Lost & Found Platform – Item Reporting System**

## **1. Introduction**

In Ethiopian cities and universities, people frequently lose personal items such as ID cards, wallets, student badges, phones, and important documents. When something is lost, there is no organized digital system to report it. Most people rely on Telegram groups, Facebook posts, or asking around.

This project proposes a simple web-based Lost & Found platform that allows users to report lost and found items in one centralized system.

## **2. Statement of the Problem**

Currently:

- Lost items are reported randomly on social media.
- There is no centralized database to store lost and found information.
- People must search manually through many posts.
- There is no structured way to match lost and found items.

Because of this, recovery of lost items is slow and inefficient.

## **3. Objectives**

The objective of this system is to:

- Develop a simple web platform where users can report lost items.
- Allow users to post found items.
- Store item details such as:
  - Item name
  - Location
  - Date
  - Description

- Contact information
- Provide a search feature to help users find matching items.
- Display recent lost and found reports in an organized list.
- Filter by location or item type (e.g., ID card, phone, wallet).

#### **4. Significance**

This system:

- Improves the chances of recovering lost items.
- Provides an organized and reliable reporting platform.
- Reduces dependence on scattered social media posts.
- Can first be implemented at university level and later expanded to city level.